

Jack White

West Lafayette, IN • (415) 847-0063 • white942@purdue.edu • LinkedIn • GitHub

Education

Purdue University

West Lafayette, IN

B.S. Computer Science and B.S. Artificial Intelligence

2023 – 2027

- GPA: 3.6/4.0
- Relevant Coursework: Introduction to Robotics, Imitation Learning for Robotics (current), Data Mining and Machine Learning, Analysis of Algorithms, Systems Programming, Probability, Operating Systems

Experience

BPM Microsystems

Houston, TX

Automation / AI Software Engineering Intern

Jul–Aug 2025; May–Aug 2026

- Built **AlgoBuilder**, an autonomous engineering workflow that develops and validates semiconductor programming algorithms by orchestrating AI agents, Salesforce/MCP tools, BPWin REST/WebSocket APIs, automated hardware QA, persistent job state, and human hardware handoffs.
- Built **PDR Builder**, an AI-assisted datasheet-to-design pipeline that extracts semiconductor pin, protocol, and electrical requirements, validates them deterministically, routes uncertain results through human review, and generates production .pdr and Excel engineering artifacts.
- Validated PDR automation on a historical corpus: **27/27 Tier-C designs passed with 0.09 noise errors/design** and zero silent errors in evaluated pin-extraction benchmarks.
- Responsible for the **BPWin REST API 1.2–1.9** upgrade for unattended hardware operation, including job observability, remote dialog control, device/socket discovery, NAND-range support, algorithm database reindexing, and capability validation.
- Previously modernized BPM's internal AI assistant, **Chip**, around Model Context Protocol; developed system/unit tests, integrated Salesforce and Google Drive tools, and reduced model-call costs.

Purdue University – Prof. Yung-Hsiang Lu

West Lafayette, IN

Computer Vision Research

Aug 2025 – Dec 2025

- Implemented computer-vision methods for piano-performance transcription to MIDI to evaluate generative systems synthesizing performance video from sheet music.

Robotics Projects

ControVirtual – XR Robotic Teleoperation and Robot Learning

Apr 2026

- Built a robotics integration stack connecting a **Meta Quest** Unity application, spoken commands, a Python WebSocket bridge, live OpenCV camera streaming, robot telemetry, and a browser-based digital twin.
- Fine-tuned the **SmolVLA vision-language-action model** on **50 teleoperated episodes** of an SO-101 robot-arm manipulation task and developed inference scaffolding to combine natural-language instructions, visual observations, and robot joint state.
- Implemented runnable voice-command, camera, digital-twin, and simulated telemetry subsystems; developed the architecture for connecting learned-policy outputs to physical robot control.
- Received **Honorable Mention – Best Use of AMD Tech at StarkHacks 2026** at Purdue University.

Technical Skills

Programming: Python, C, C++, C#, Java, TypeScript/JavaScript, SQL, Assembly

Robotics / Computer Vision: PyTorch, OpenCV, Unity, Meta XR, LeRobot, SmolVLA, vision-language-action models

AI / ML: Hugging Face, LLM APIs, Model Context Protocol (MCP), pandas, NumPy

Systems / Infrastructure: Linux, REST APIs, WebSockets, FastAPI, Flask, Docker, Git, Mercurial

Hardware: Semiconductor device programming, NOR/NAND flash, microcontrollers, programmer hardware